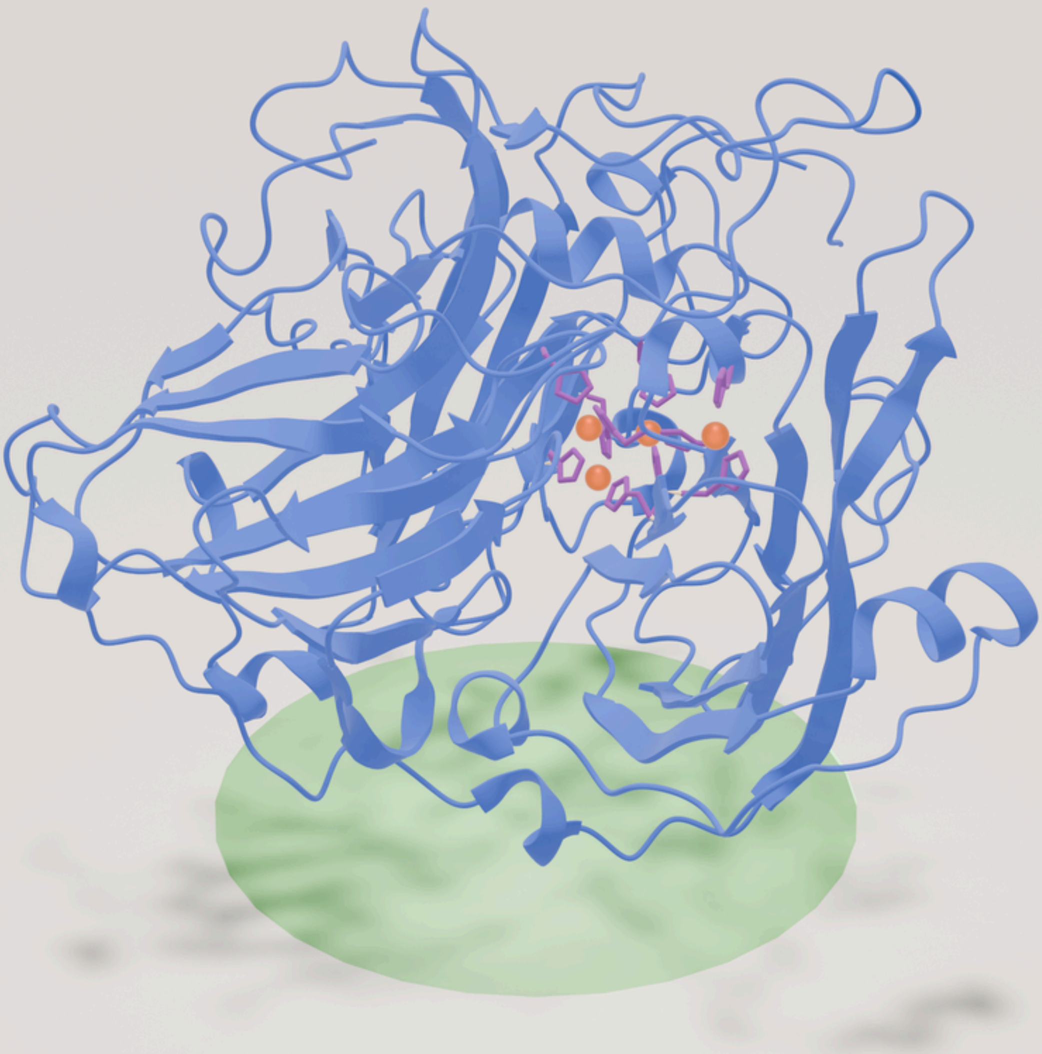
Facultat de Biociències

An *in silico* analysis of laccase thermostability determinants for the improvement of rational enzyme engineering



Author: Adrià Salarich Soldevila

Degree in Microbiology

Tutor: Eloi Parladé Molist, PhD

Abstract

Laccases (EC:1.10.3.2) are enzymes of the multi-copper oxidase family capable of degrading a wide range of substrates using oxygen as the final electron acceptor and producing water as a by-product of the reaction. Owing to this characteristic, they are promising tools in the field of bioremediation. However, the application of laccases in large-scale procedures is hindered by the low thermostability of many of these enzymes, a property whose molecular determinants remain to be fully characterised. Therefore, the aim of this project was to analyse thermostable and non-thermostable laccases to elucidate their physicochemical differences and identify key residues contributing to thermostability. To achieve this, multiple-sequence alignments of thermostable and non-thermostable laccases were constructed and analysed through statistical tests and the Kullback-Leibler divergence. The analysis revealed a high degree of global similarity between the two groups of enzymes, suggesting that thermostability is mostly determined by local conservation patterns. Using the Kullback-Leibler divergence and *Bacillus subtilis* CotA structure as a reference, several discriminating positions were identified and structurally interpreted, revealing conservation patterns with a potential to be used in the targeted rational design of laccases to increase their thermostability.

Keywords: bacterial laccase, thermostability, bioremediation, rational engineering, multiple-sequence alignment (MSA).

Table of contents

Introduction

Laccases (EC:1.10.3.2) are enzymes belonging to the multicopper oxidase (MCO) family, capable of binding to and degrading a wide range of phenolic and non-phenolic compounds, and even some inorganic compounds,[1] using oxygen as the final electron acceptor and producing water as a by-product of the reaction. The electron transfer is possible thanks to the four copper atoms which form the active center of the enzyme.[2,3] Laccases are widely distributed in nature. They play important roles in plants, insects, fungi and bacteria. However, bacterial and fungal laccases are more appealing in the field of bioremediation due to their distinctive properties.[2,4] Specifically, bacterial laccases are generally regarded and upheld for their higher stability against environmental conditions, which makes them ideal candidates for some industrial and environmental applications.[2]

These enzymes belong to the cupredoxin superfamily of enzymes, characterised by the cupredoxin fold, in which two β-sheets are arranged into a Greek-key barrel. Laccases usually contain three cupredoxin domains, but two-domain laccases also exist (mainly in bacteria) and are active in a homotrimeric form.[2,3] The active center of laccases comprises four copper atoms (see Figure 1) – termed Type 1 (T1), Type 2 (T2) and Type 3 (T3) according to their configuration – which are distributed in two separate copper sites: a mononuclear site, where the T1 copper ion is located, and a trinuclear site containing the remaining 3 copper ions, where one T2 and two T3 (T3 and T3') copper ions form a cluster.[2] The geometries of the copper sites are conserved among different laccases.[3] The T1 copper is coordinated by two histidine (His) residues and one cysteine (Cys) residue as equatorial ligands, and a variable axial ligand. In the trinuclear site, eight total His residues coordinate the T2 and T3 copper ions.[2]

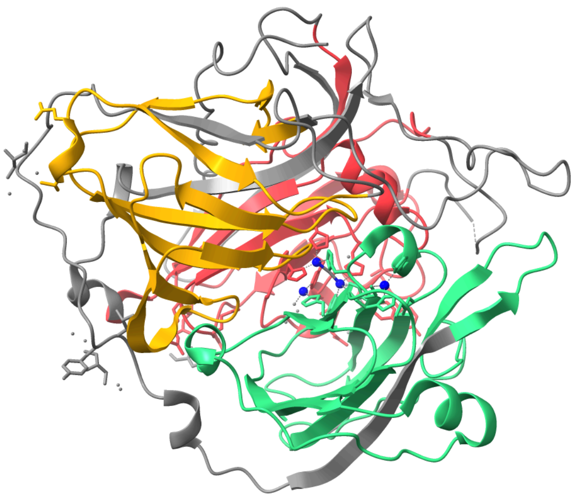


Figure 1. The spore-coating laccase of *Bacillus subtilis*, CotA. Structure obtained from PDB (1GSK). The domains are colored according to PFAM:[5,6] Domain 1 is colored in red; Domain 2 is colored in yellow; and Domain 3 is colored in green. The copper atoms are colored in dark blue. Gray regions are uncharacterised.

Current state of laccase application in bioremediation

By virtue of their wide range of substrates, laccases are promising enzymes in the field of bioremediation as a green alternative to more aggressive and less sustainable chemical processes, with potential applications in the degradation of pesticides, textile dyes, antibiotics and others.[1,2,7] Many experiments have demonstrated the efficacy of laccases (mainly of bacterial and fungal origin) in the removal or degradation of various contaminants including:[1,2] antibiotics,[8] textile dyes such as malachite green,[9] and pesticides.[10] Bacterial laccases in particular have been notoriously successful in the bioremediation of various industrial dyes from water.[2] Currently, a notable amount of well-characterised laccases are commercially available for all sorts of industrial purposes as well.[11]

Despite the potential of laccases, many authors express challenges that must be overcome for wider adoption in large-scale bioremediation procedures. Factors such as temperature, medium pH, salinity or the presence of organic solvents in the operation medium can severely hinder enzyme stability and impair the bioremediation process.[2] For these reasons, the use of free laccase seems less ideal at large scales, with Paul et al. describing the commercial application of free enzyme as not feasible.[12]

Immobilization is a promising tool in many enzyme-based mass operations, including bioremediation. Thanks to these systems, enzyme stability and, therefore, reusability, is greatly augmented.[13] However, complexity and cost of production remain high, and more testing in real environmental conditions is needed.[12,14] The engineering of laccases is also an appealing method to improving resistance to environmental conditions. In this case, scientists face a gap of knowledge in the prediction of the effect of specific mutations and current methods have a hard time maintaining both stability and function simultaneously.[2]

Thermostability importance and determinants

Although bacterial laccases present higher resistance in comparison to those of other organisms, the poor thermostability of most of these enzymes remains a limitation. Enzyme reusability, an important determinant of the viability of any enzyme-based process, is notably constrained by this property.[2,13,15] Additionally, high-temperature processes are generally desired in large-scale applications due to increased reaction rates and reduced need for cooling, among other advantages.[16]

Many elements play a role in determining a protein's thermostability. Amino acid composition of solvent-accessible (SA) and low-solvent accessibility positions is a notable factor. Solvent-accessibility is generally measured by a residue's Relative Solvent Accessibility (RSA), which can be calculated using the following formula:[17]

$$RSA = ASA/MaxASA$$

Where ASA is the Solvent-Accessible Surface Area (also expressed as SASA) of the residue and $MaxASA$ is its maximum possible ASA, both frequently measured in $Å^2$. In the literature, SA residues are classified as those exceeding a 20% RSA value, whereas residues are considered of low-solvent accessibility if their RSA is below the 20% threshold.[18]

For low-solvent accessibility regions in particular, hydrophobicity is a major contributor to thermostability and stability in general,[19] as well as to the protein folding process.[20] Comparisons between thermophilic and mesophilic proteins have demonstrated that the preservation of core hydrophobicity is an essential factor in determining a protein's thermostability. Mutations to a protein's hydrophobic core have also been reported to greatly alter protein stability, with amino acid size changes resulting in instability.[19] Even though the exact mechanisms through which hydrophobicity favours stability are still to be determined, it has been shown that the presence of hydrophobic residues in proteins helps fill cavities thanks to their large side-chains,[21] as well as to maintain hydrophobic-hydrophobic interactions and therefore stability.[19]

In SA regions, higher numbers of exposed salt-bridges and increased presence of charged residues have been observed in many thermostable proteins. The latter are thought to favour stability through electrostatic stabilisation with water.[22]

In general terms, the enrichment of certain amino acids or amino acid classes seems to play a role in thermostability. Hydrophobic and larger amino acids are generally enriched in thermostable proteins. Arginine (Arg), threonine (Thr), alanine (Ala) and glutamic acid (Glu) contribute to thermostability through the formation of long-range interactions.[16] Contrarily, methionine (Met) and uncharged polar amino acids are less frequent. Charged amino acids are seemingly preferred;[23] it is believed that these amino acids are favoured thanks to their ability to form salt bridges, which help stabilise the protein's structure in high temperatures. Proline (Pro), which is considered to have a rigid backbone, has been demonstrated to have a higher frequency in thermophilic proteins. It has been suggested that its rigidity helps decrease entropy of unfolding, especially when located on turns or N caps of helices. Several amino acid patterns have also been suggested to be more abundant in thermophilic proteins.[19] Finally, higher presence of disulfide bonds is expected in thermostable enzymes.[22]

Research specifically focusing on laccases has pointed to glutamine (Gln), cysteine (Cys), Thr, and phenylalanine (Phe) content, and the presence of hydrophilic and acidic amino acids, to be important determinants of their thermostability.[16] Pro enrichment has also been described as a stabilizing factor of the thermostable spore-coating laccase CotA from *Bacillus subtilis*.[1]

Despite the current knowledge presented, protein thermostability prediction is a very complex process with many more factors at play than those mentioned, which traditional experimental methodologies fail to properly analyse and identify[16]. Thermostability determinants are not yet fully understood,[23] and *in silico* approaches are a promising tool for bridging this knowledge gap.[16]

Hypothesis and objectives

While enzyme immobilization is a promising technique to enhance laccase thermostability and thereby support their use in bioremediation, currently, this technology faces some drawbacks that highlight the need to obtain enzymes with higher intrinsic thermostability. Although many bacterial laccases that possess this property are known, features favouring their thermostable nature are still not completely understood. Therefore, identifying the differences between thermostable and non-thermostable laccases (hereafter referred to as generic laccases) may aid in the discovery of new bioremediation candidates and guide the rational engineering of more stable enzymes.

For this project, it is hypothesised that the study of generic and thermostable bacterial laccases through MSA analysis will reveal important thermostability determinants. Specifically, low-solvent accessibility positions are expected to show increased hydrophobicity and enrichment in apolar residues, whereas SA positions are expected to show greater enrichment in charged and proline residues, as well as residues capable of forming disulfide bonds. Finally, thermostable laccases are expected to exhibit amino acid substitutions favouring stability.

Therefore, the present work aims to form a better understanding of the physicochemical properties that are shared among thermostable bacterial laccases through the use of multiple sequence alignment (MSA) analysis methods, while considering structural context to determine the impact of the features. More specifically, the physicochemical properties of low-solvent accessibility and SA residues, as well as the amino acid enrichment of these regions in thermostable and generic laccases, will be explored and compared through statistical analyses. The study will seek to identify amino acid substitutions that can potentially increase the thermostability of known laccases through the use of enzyme engineering strategies, as well as to define characteristics to be targeted in newly discovered enzymes.

Methodology

Sequence search and collection

Generic bacterial laccase sequences were retrieved from UniProt[6] and NCBI[24] Protein databases using advanced filtering. Only laccases between 400-700 amino acids with EC: 1.10.3.2 annotation were selected to preserve dataset quality and avoid small laccases or partial records. Entries with "predicted", "uncertain" (UniProt), "partial" and "putative" (NCBI) keywords were excluded.

Thermostable bacterial laccases were identified from BRENDA[25] (EC:1.10.3.2) using filters "stable at" or "optimal temperature" >50ºC. This is a relatively low threshold in comparison to those frequently used in protein thermostability studies,[26] chosen specifically to maximise dataset size given the low expected number of characterised thermostable laccases. The resulting BRENDA entries were cross-searched in UniProt, if available, using the Accession Codes. Additional sequences were added to the dataset through targeted UniProt and NCBI searches, using keywords such as "thermostable" or "thermoresistant". A reduced number of proteins reported in the literature with available sequences were also collected.

Processing of the datasets

To reduce the overrepresentation of *Bacillus* sequences, proteins were clustered with CD-HIT (v4.8.1)[27] to obtain a smaller set of representative laccases. The generic dataset was clustered at an identity cutoff of 90%. Due to its reduced size, the thermostable set was clustered at a less strict 95% identity cutoff, and analyzed in parallel with the non-clustered dataset in some steps. After clustering, CD-HIT-2D (part of CD-HIT) was used to check for 100% identical sequences between the thermostable and generic datasets, ensuring no sequence contributed to both datasets simultaneously.

Multiple sequence alignment and curation

The generic and thermostable datasets were aligned using the MAFFT web server (v7.526)[28] at default parameters. Subsequently, the alignments were trimmed with trimAl (v1.5.1)[29] to facilitate downstream analysis. Three additional alignments were generated by adding the sequence of *Bacillus subtilis* CotA (Uniprot: P07788) to all the previous alignments using MAFFT's `--add` function. These were also ultimately trimmed with trimAl. P07788 was consequentially trimmed along with all other sequences in each alignment.

Alignment analysis

Prior to all analyses, RSA values for all residues in the *B. subtilis* CotA structure (PDB: 1GSK)[30] from PDB[31] were calculated using FreeSASA (v2.1.2).[32] Residues were classified as SA (> 20% RSA) or low-solvent accessibility (< 20% RSA). Then, they were located in ChimeraX[33] and exported to text files with PDB sequence numbering. Position Specific Scoring Matrices (PSSMs) were later generated from the trimmed MSAs containing P07788 using Goalign (v0.4.0).[34] Residues with RSA < 20% are hereafter referred to as buried for simplicity, acknowledging that some of these positions may still retain partial solvent accessibility despite falling below the conventional threshold.

Using P07788 as a mapping reference, the PDB numbers of buried and SA residues were matched to their respective PSSM rows. Positions removed during the earlier trimming steps were avoided. Separate PSSMs containing only SA or buried residues were obtained for comparative analysis. All MSAs, before and after processing, were visualised using Jalview (v2.11.5.1).[35]

Three complementary analyses were applied to the datasets and are described in detail in the following sections. All analyses were implemented in Python 3 with the aid of Claude Sonnet 4.6,[36] using packages SciPy, NumPy and pandas.[37–39] Matplotlib[40] was also used to create alignment plots. The scripts and output files were thoroughly manually inspected before use.

Amino acid enrichment

To compare the amino acid composition of SA and buried residues respectively between generic and thermostable laccases, the 20 amino acids were classified into functional groups according to IMGT standardised criteria.[41] For each group, frequencies were summed across all positions and normalised accounting for alignment gaps, creating contingency tables comparing the frequency of the amino acids in the group *vs.* all other amino acids. Then, Chi-squared (χ^2) tests of independence were performed on each contingency table to determine whether amino acid group frequencies were significantly different between thermostable and generic laccases.

Physicochemical properties

The physicochemical properties of the thermostable and generic laccases under study were also analyzed. The properties tested and the scales used can be found in Table 1.

Physicochemical property	Scale
Hydrophobicity	Kyte-Doolittle[42]
Volume	A. A. Zamyatnin[43]
Charge	IMGT[41]*
Polarity	Zimmerman[44]
Flexibility	Bhaskaran R., Ponnuswamy P.K.[45]**

Table 1. Physicochemical property scales used in the analysis. *Charge classification based on the amino acids' formal charge at pH = 7 was extracted from IMGT's standardised criteria[41] for consistency with the amino acid enrichment analysis. **The index values were rounded down for the analysis following ExPASy Server[46] ProtScale values.

The frequency-weighted average value for each physicochemical scale was calculated at every position in the alignment, accounting for the relative frequency of each amino acid at that position. Then, the position-specific property values were subjected to comparison-of-means statistical tests. Additionally, the average value for each scale was calculated across all positions to obtain global summaries. Since normality of the data distribution is uncertain, both Student's T-test and Mann-Whitney U were calculated to ensure robust statistical inference, using the latter as the main reference.

Kullback-Leibler (KL) divergence test

The identification of the most discriminating positions between thermostable and generic laccases was carried out using the KL divergence. For each buried and exposed position common in both datasets, KL divergence was calculated in both directions using the formula:

$$D_{KL}(P||Q) = \sum_{x \in X} P(x) \cdot ln\frac{P(x)}{Q(x)}$$

Where P and Q are the amino acid frequency distributions of the PSSMs corresponding to each common position, and x represents each of the 20 standard amino acids. A symmetric measure was obtained from the average of both values:

$$\frac{D_{KL}(P||Q)+D_{KL}(Q||P)}{2}$$

Prior to calculation, a pseudocount of 0.001 was added to all frequencies to handle zero values. Subsequently, the data were re-normalised to ensure frequencies added up to one, accounting for the exclusion of alignment gaps as well.

Positions were ranked by KL divergence value to identify the most discriminating locations. Shannon entropy (SE) was also calculated at each position from its respective amino acid distribution to measure within-group variation. ΔSE for each position was calculated by subtracting the generic SE value from the thermostable SE value *($ThermoSE_n$ - $GenericSE_n$)*. Mean SE values were calculated by averaging SE across all common positions.

Results

Number of sequences obtained and curation steps

The different search methods used allowed for the retrieval of 164 generic laccase sequences and 59 thermostable proteins. After clustering, the generic dataset was reduced to a more representative 77 sequences, whereas the clustering applied on the thermostable dataset reduced the number of sequences down to 18. For downstream analysis, it should be noted that in the generic alignment, 45 residues of P07788 were trimmed by trimAl, likely due to the increased sequence diversity present in the set, whereas only 12 residues were lost in the non-clustered thermostable set, and only 3 were trimmed from the 95%-clustered dataset MSA. For this reason, the 95%-clustered thermostable alignment was used for all analyses going forward.

Comparative analysis of thermostable and generic laccases

Number of buried and solvent-accessible positions

In P07788, 278 residues were identified as buried (<20% RSA) and 224 were classified as exposed (>20% RSA) using 1GSK as a structural reference, accounting for 502 of the 513 residues in P07788; the remaining 11 residues lack coordinates in 1GSK due to structural incompleteness. From the generic PSSM, 267 buried positions and 197 exposed positions were successfully extracted, indicating that trimAl removed 11 buried amino acids and 27 SA amino acids from P07788. In contrast, all 224 and 278 exposed and buried positions were extracted from the 95%-clustered thermostable PSSM.

After this process, 197 SA and 267 buried positions in common between 95%-clustered and generic MSAs were left available for comparison.

Amino acid composition and physicochemical property analysis of solvent-exposed and buried residues

The amino acid enrichment analysis revealed no significant enrichment of any of the defined amino acid groups in either buried or SA positions.

The analysis of the physicochemical properties of the enzymes revealed subtle differences; laccases in the thermostable dataset were found to be less hydrophobic at their SA positions (MW $p = 0.023$) than those classified as generic. Additionally, the buried positions of thermostable laccases had a lower polarity value (MW $p = 0.013$). On the other hand, t-tests did not return any significant result.

KL divergence results

For the KL divergence test, 267 buried and 197 SA positions were tested. Summarised results for this test can be found in Table 2. For any particular position, the symmetric KL divergence indicates the level of discrimination between the two laccase groups. In contrast, the Shannon entropy value represents within-dataset variability. Thus, a positive entropy difference for a given position indicates that variability is higher in thermostable laccases, whereas a negative entropy difference means higher variability in generic laccases.

Among SA positions, 125 in 1GSK shows the highest KL divergence score and a negative ΔSE value, indicating that the amino acid frequencies in this position differ substantially between the two datasets, and that the position is conserved in thermostable laccases while being less conserved in the generic group. Among buried residues, position 333 presents a similar case, with a more negative ΔSE value representing a more pronounced difference in conservation. Thermostable laccases showed a lower mean entropy for both residue classifications as exemplified by the ΔmeanSE values in Table 2.

Buried residues			SA residues		
1GSK residue number	**Symmetric KL divergence**	**ΔSE**	**1GSK residue number**	**Symmetric KL divergence**	**ΔSE**
333	2.470	-0.963	125	2.609	-0.704
322	2.287	-0.937	5	2.431	0.046
100	2.140	-0.8	137	2.345	-1.095
123	2.098	-0.676	378	2.300	-0.436
271	2.082	-0.42	319	2.233	-0.137
335	2.067	-0.18	44	2.169	-2.262
127	2.029	-1.788	484	2.165	-1.311
101	1.870	-1.263	213	2.073	-0.584
320	1.850	-0.467	447	1.966	-1.432
220	1.743	-1.511	4	1.963	-1.193
385	1.551	-1.098	328	1.955	-1.488
332	1.515	-1.304	446	1.926	-1.523
219	1.487	-0.663	136	1.823	-0.945
399	1.393	-0.468	138	1.809	-0.807
318	1.378	-1.188	449	1.802	-1.086
240	1.356	-1.123	384	1.774	-1.036
142	1.344	-0.715	369	1.770	-0.707
505	1.320	-1.242	221	1.770	-1.294
43	1.311	-1.069	323	1.769	0.097
295	1.295	-1.101	216	1.741	-0.927
ΔmeanSE		-0,442	**ΔmeanSE**		-0,582

Table 2. Summarised KL divergence test results showing the top 20 most discriminating positions ordered by symmetric KL divergence score. In contrast, ΔmeanSE was calculated across all common positions.

Discussion

Taxonomic bias and alignment diversity in the thermostable and generic datasets

Even though the clustering applied to the thermostable data was less strict, this step still produced a small thermostable dataset, likely due to the aforementioned overrepresentation of *Bacillus* laccases. This supposition is supported by the formation of two notably large clusters, each containing 16 proteins. The first consisted mainly of *B. subtilis* laccases, whereas the second was largely made up of *Bacillus pumilus* laccases; overall, both contained only *Bacillus* sequences. Despite this increased relative abundance, all sequences collected met the thermostability criteria and thus should remain representative of thermostable features.

The trimming step supported the suspected differences in diversity between both sets; the generic dataset contained many more gaps, likely due to the high sequence diversity present in the set. The gaps in the generic alignment were appropriately handled by trimAl, drastically reducing the number of columns, as can be observed in Figure 2. As expected given the narrower taxonomic and functional range in the 95%-clustered dataset, its alignment was much more homogenous in comparison to its counterpart, but gaps were still abundant, warranting the trimming step. Overall, this procedure was considered necessary to prevent the presence of alignment artifacts, such as excessively gappy columns, potentially affecting downstream analysis.

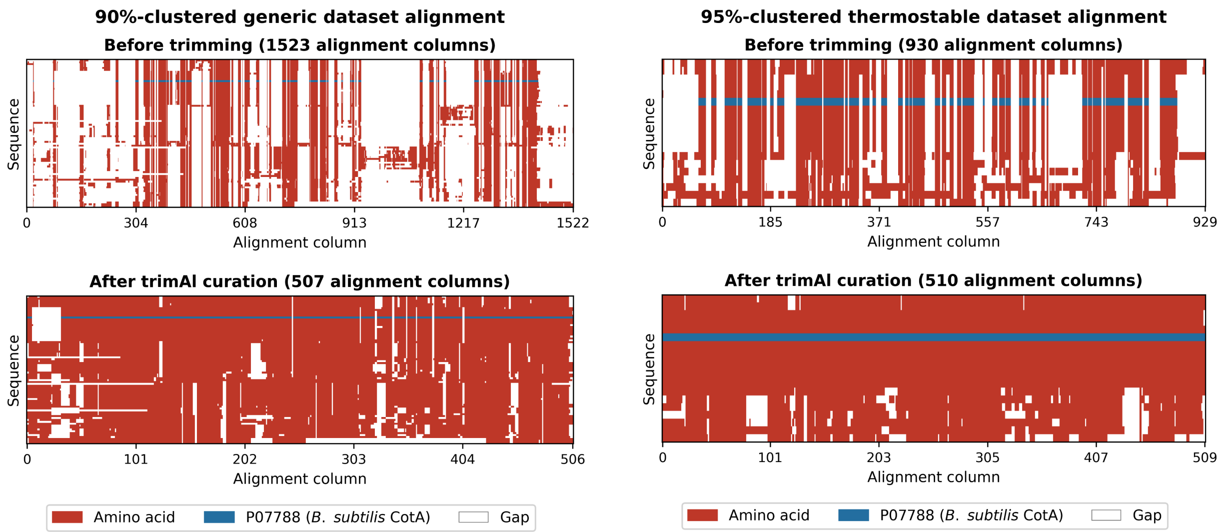


Figure 2. Comparison of MSA quality before and after trimming. P07788, the spore-coating laccase of *B. subtilis*, is highlighted in blue. Both generic and thermostable datasets are shown before and after trimming.

Thermostable and generic laccases are similar at a global scale

The combined results of the amino acid group enrichment and physicochemical tests are indicative that there is very little global difference between thermostable and generic laccases in the data, contrary to what was expected. In the amino acid group enrichment analysis, no amino acid group was found to be enriched at a statistically significant level in any buried or SA position. Notably, thermostable laccases were found to be less polar at their buried positions and less hydrophobic at their surface positions. Given the inverse correlation between polarity and hydrophobicity, these findings fit the classical view of thermostable enzymes being more hydrophobic at their cores and more hydrophilic at their surfaces.[19,20] However, it should be noted that the t-test did not corroborate these results, suggesting a modest effect. Overall, it remains clear that the global differences in buried and SA residues between thermostable and generic laccases are small.

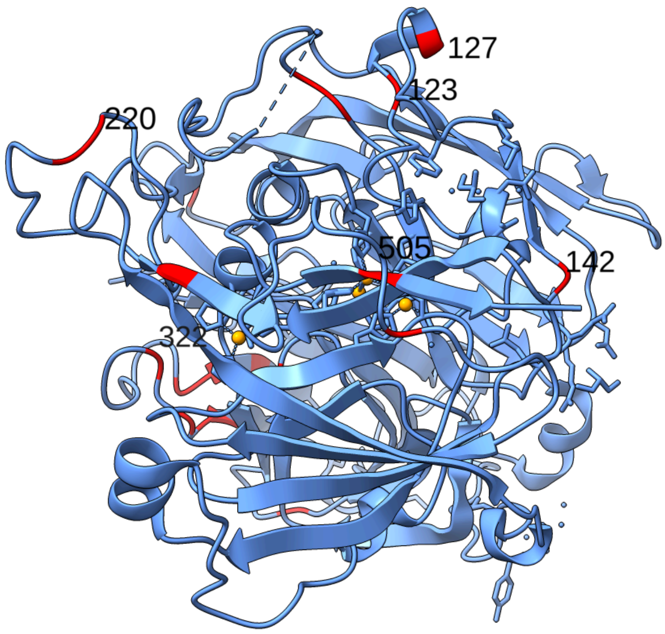
Position-specific amino acid conservation in thermostable laccases

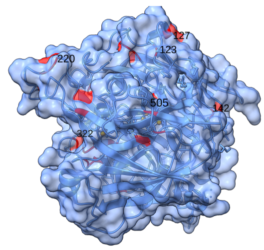
As indicated in the methodology section, the 20% RSA threshold used to classify *B. subtilis* CotA residues into buried and SA does not produce discrete groups. Rather, solvent accessibility is a gradient for which the 20% threshold is a conventional cutoff. Thus, positions classified as buried in this work may still retain partial solvent accessibility. This distinction between the conventional classification and the nature of solvent accessibility is particularly relevant when interpreting the KL divergence results. Notably, all of the top 20 most discriminating buried positions (see Table 2) except position 123 appear to be at least partially exposed to the solvent (see Figure 3), indicating that truly core residues may be more conserved across thermostable and generic laccases than partially exposed positions, thereby ranking lower in KL divergence.

In addition to the general conservation of truly buried residues, an additional structural pattern is evident from the KL divergence results. In both buried and SA positions, almost all of the top 20 most discriminating positions fall on loops and at the termini of beta-sheets and alpha-helixes (see Figures 3 and 8). It is known that loop regions are generally more flexible in proteins,[26] and it has been shown that their rigidification can increase thermostability in enzymes and in laccases specifically.[21,47] The conservation of specific residues at these positions in thermostable laccases therefore suggests selective pressure to rigidify these flexible regions, potentially raising the energetic cost of unfolding. The following sections discuss the most notable of these positions in detail.

Structural analysis of highly-discriminating buried positions

Three buried positions containing proline (see Figure 3) were found to be more conserved in thermostable laccases than in generic laccases (see Table 2). This finding is consistent with the role of proline residues in general enzyme stability, increasing backbone rigidity, as well as with proline content being a stabilising factor in CotA.[1] Notably, P220 and P142 are located on turns, where their structural function is most well-understood.[19] In contrast, P505 is located amidst a beta-strand at the edge of a beta-sheet in the third domain of the protein. Despite this amino acid being generally thought of as a beta strand breaker, research shows that Pro residues can be desirable at edge positions of beta-sheets, where they may help limit beta interactions and prevent protein aggregation, something that beta-sheets are prone to given the availability of hydrogen bond-forming residues at their edges.[48,49]





322	Cys
123	Phe
127	Phe
220	Pro
142	Pro
505	Pro

Figure 3. Top 20 most discriminating buried residues mapped on 1GSK. The amino acid corresponding to each of the six labeled positions can be seen on the table.

Although C322 ranked second in KL divergence (see Table 2), no obvious stabilising interactions involving this position were observed. Despite the role of cysteine in copper coordination in laccases[3] and other enzymes,[50] C322 does not seem to participate in any interaction of this kind. Rather, this position seems to contact a nearby Arg residue in 1GSK, perhaps forming a local stabilising interaction, but the overall functional role of this conservation in thermostable laccases remains unclear.

The top ranked buried position, N333, was located at the N-terminus of a beta-strand, preceded by a highly discriminating position, A332, and neighbouring another highly-discriminating position, N318 (see Figure 4.A and Table 2). All of these are conserved in thermostable laccases, with position N333 displaying great selectivity for asparagine (see Figure 5.A). Additionally, it is worth noting that SA residue S319 is also in the top 20 most discriminating positions for SA residues, suggesting high selectivity in this area for thermostable laccases. However, the combined functional role of these four positions is less clear than previously mentioned residues, and the reason behind their conservation also remains unclear.

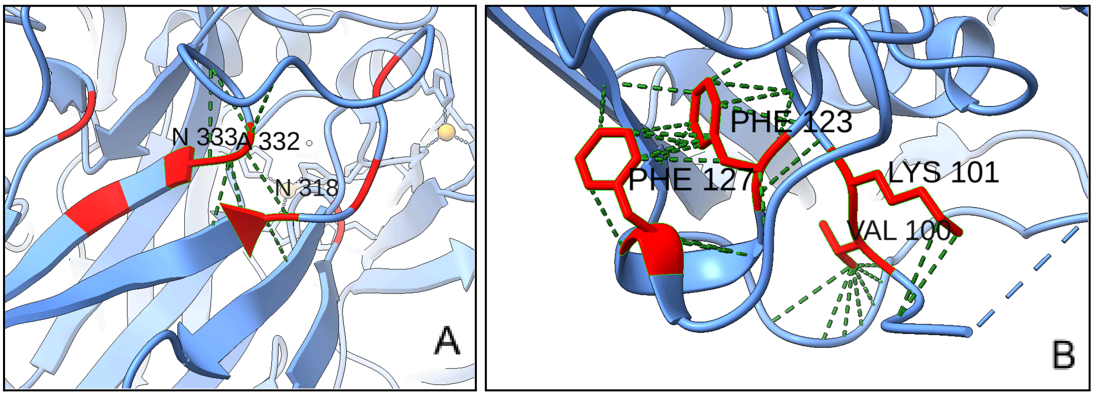


Figure 4. Representation of various highly discriminating buried residues on 1GSK. The green dashed lines represent atomic contacts as measured by ChimeraX. Representation of residues (A) N333, A332 and N318; and (B) F123, F127, V100 and K101.

Finally, residues F123, F127, V100 and K101 (see figure 4.B) were also found to be in the top 10 most discriminating positions and appear to be spatially proximal. The hydrophobic nature of Phe and Val suggest that their role in this region is the formation of a small stabilizing hydrophobic cluster.[51] Specifically, F127 is found in a small alpha-helix, whereas V100 and F123 are located in neighbouring loops. Despite being classified as buried, K101 is likely at least partially exposed to the solvent, where it may contribute to stability by forming salt bridges through its polar charged side chain.[19] It is worth noting that F127 has the lowest ΔSE among the top 20 most discriminating buried residues (see Table 2), indicating strong conservation in thermostable laccases relative to generic laccases, further demonstrated by the conservation plot in Figure 5.B.

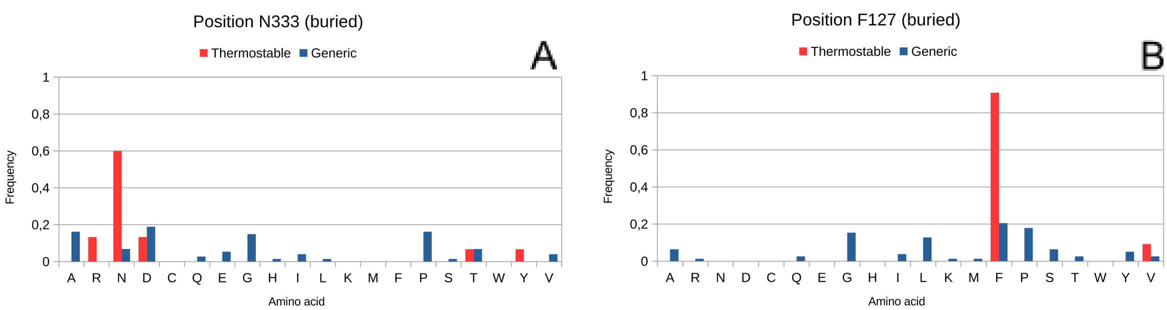
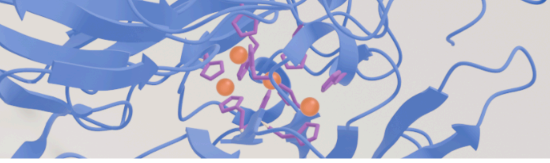


Figure 5. Amino acid frequency plots for buried positions (A) N333 and (B) F127.

Structural analysis of solvent-accessible residues

As observed for buried positions, three SA Pro residues in 1GSK, P44, P328 and P384, were found to be highly discriminating between thermostable and generic laccases. Positions P44 and P328 are found in exposed loops, where their role has already been discussed as stabilizing in thermophilic proteins.[19] P44 in particular has the lowest ΔSE value among the



top 20 most discriminating exposed positions (see Table 2), and its conservation plot (see Figure 6.A) shows it is extremely well-conserved in thermostable laccases while being much more diverse in generic laccases. However, the role of P384 seems to be different due to its structural positioning (see Figure 7.A). This residue is located at the N-terminal cap of the second beta-strand in a beta-hairpin motif. Additionally, it is in contact with Q378, another highly discriminating position which is at the C-terminal cap of the first beta-strand of the motif. In this context, the co-conservation of these two residues could indicate a coordinated stabilizing role through backbone rigidification by P384 and side-chain hydrogen bonding by Q378.[52]

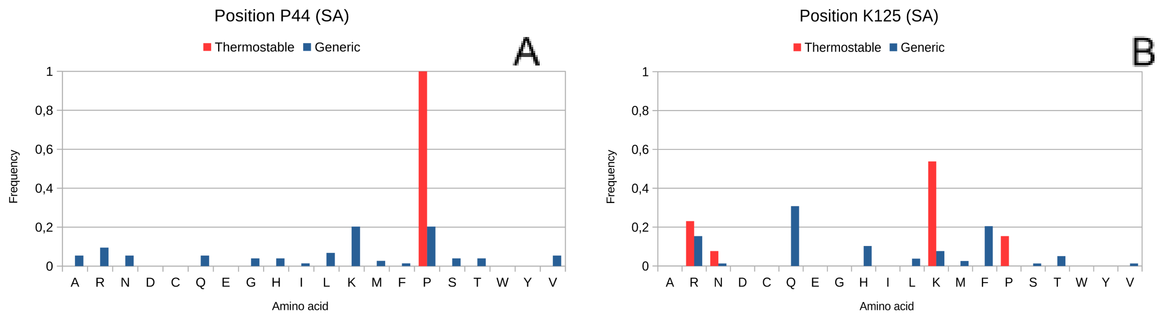


Figure 6. Amino acid frequency plots for SA positions (A) P44 and (B) K125.

E446, L447 and Y449 are co-located in a prominent surface loop (see Figure 7.B). Despite E446's role being less clear, L447 and Y449 appear to participate in a side-chain interaction which could play a part in stabilizing the loop.

Positions R136, E137, and V138 all rank high in KL divergence. R136 and E137 are part of a loop preceding an alpha-helix capped by V138 (see Figure 7.C). Although their precise structural interaction is hard to interpret, the co-conservation of these 3 residues could be indicative of strong selective pressure to maintain the structure of this region.

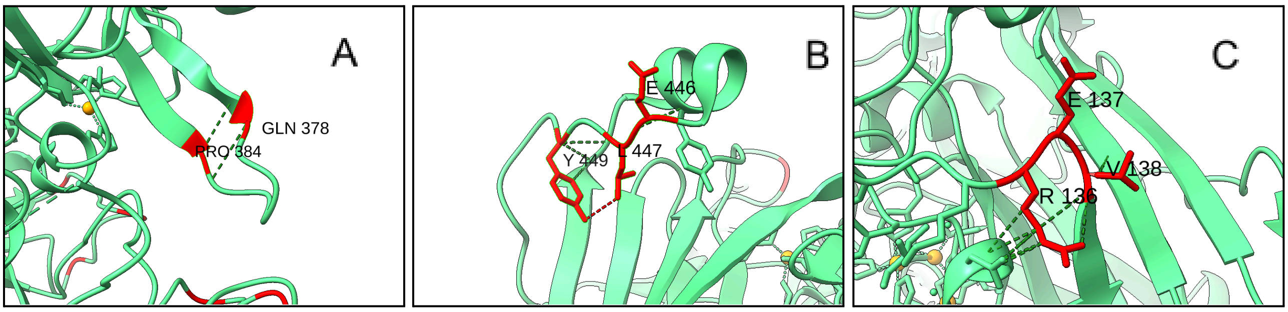
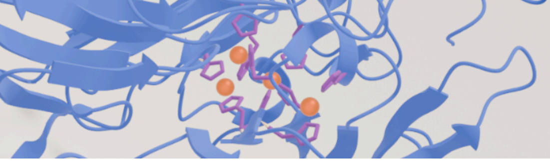
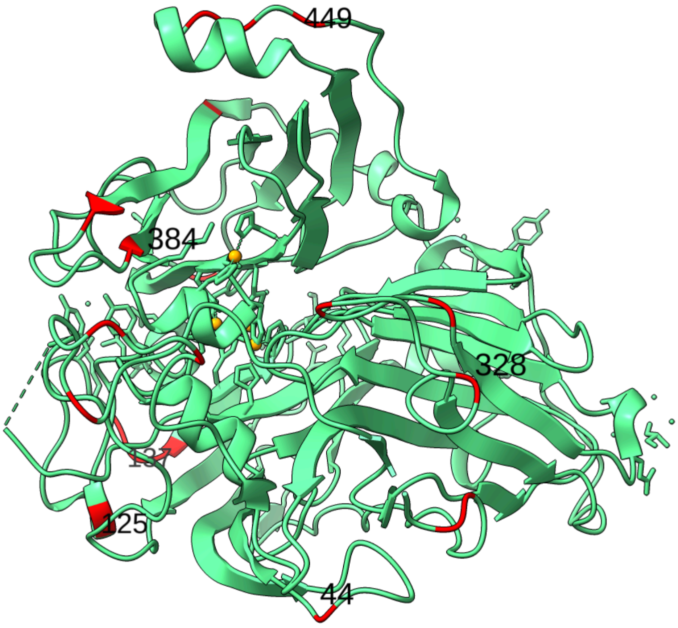
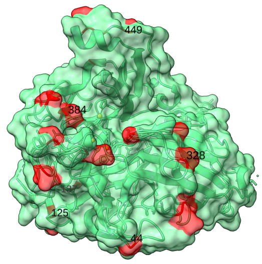


Figure 7. Representation of various highly discriminating SA residues on 1GSK. The green dashed lines represent atomic contacts as measured by ChimeraX. Representation of residues (A) P384 and Q378; (B) E446, L447 and Y449; and (C) R136, E137, and V138.



Lastly, position K125, the top ranked discriminating position among SA residues, is part of an alpha-helix and seems to be interacting with multiple residues of a neighbouring loop, potentially stabilising the loop-helix connection. This position is also very well conserved in thermostable laccases relative to generic laccases (see Figure 6.B).





125	Lys
137	Glu
44	Pro
328	Pro
449	Tyr
384	Pro

Figure 8. Top 20 most discriminating SA residues mapped on 1GSK. The amino acid corresponding to each of the six labeled positions can be seen on the table.

Conclusions

Laccases are enzymes of the MCO family with considerable potential in the field of bioremediation thanks to the capacity of these enzymes to efficiently degrade a wide range of organic contaminants. However, the low thermostability of laccases is a limitation to their application in large-scale bioremediation procedures. Despite the characterization of many thermostable laccases, the features underlying their stability are yet to be fully understood. Thus, the aim of the present work was to expand the knowledge on the differences between thermostable and generic bacterial laccases to form a better understanding of this property, and guide the rational design and discovery of laccases to facilitate their application in the field of bioremediation.

To achieve this, multiple sequence alignments of thermostable and generic laccases were constructed and analysed through amino acid group enrichment and physicochemical property analyses, as well as KL divergence.

In combination, the amino acid enrichment and physicochemical property analyses revealed a high degree of global similarity between thermostable and generic laccases in the dataset, suggesting that their differences lie mainly in local conservation patterns rather than global structural changes. This interpretation was supported by the results of the Kullback-Leibler divergence, which identified positions with different conservation patterns across both datasets, the majority of which fall on loops and secondary structure termini, consistent with a stabilisation strategy based on the rigidification of exposed or partially-exposed flexible regions. Thermostable laccases showed lower mean Shannon entropy across both solvent-accessible (ΔmeanSE = −0.582) and buried positions (ΔmeanSE = −0.442), indicating higher overall conservation.

Among the most notable findings, P44, located on a solvent-accessible (SA) loop, showed complete proline conservation in thermostable laccases, whereas this position was much more variable in the generic group, representing the most conserved individual position identified. Additional proline conservation in other buried and SA positions in a similar structural context supports the rigidification of flexible regions as a thermostabilisation strategy within the group. A putative partially-exposed hydrophobic cluster was also found through the conservation of residues F123, F127, V100 and K101 in thermostable laccases, with a potential role in the stabilisation of this region. Another notable finding was the conservation in thermostable laccases of P384 and Q378, located in the caps of two beta-strands forming a beta-hairpin, with a potential stabilising role through backbone rigidification by P384 and side-chain hydrogen bond formation by Q378. Despite these residues having a promising role in the stabilisation of thermostable laccases, whether these conservation patterns increase thermostability when introduced in generic enzymes remains to be validated through *in vitro* studies.

Several other conservation patterns that were detected through the KL divergence, such as positions N333, N318, A332, and C322 have a less clear functional role. Therefore, future studies should be carried out focusing on elucidating their potential stabilising roles.

Overall, the analysis was able to locate highly discriminating positions between thermostable and generic laccases, with a potential to be used in the targeted rational engineering of these enzymes. Additionally, the positions identified can be employed as a search strategy for the elucidation of novel bioremediation candidates. If validated, these findings could represent a valuable contribution to the application of laccases in bioremediation procedures.

Limitations

Despite the steps taken to reduce the relative abundance of proteins of the *Bacillus* genus, their persistent overrepresentation is a factor that should be considered when extrapolating the results obtained in this project to enzymes of different origins. The use of *B. subtilis* CotA as the only reference sequence with which to map the results could further limit the generalisability of the findings and their structural interpretation. Future works using similar methodologies should focus on improving dataset curation with comprehensive database searches and systematic literature reviews. The ongoing characterisation of new laccases from more taxonomically diverse organisms could allow for more representative analyses as well.

Additionally, the criteria used to define thermostability in the database searches were based around a relatively low 50ºC threshold, which could have introduced low-thermostability enzymes in the analysis, diluting the strength of the results. This threshold was nonetheless chosen deliberately to increase dataset size in an effort to compensate for the reduced number of available thermostable laccases found in the databases at the time of the study. Indeed, this is another limitation that the characterization of new thermostable laccases could help remediate.

Finally, as mentioned in the conclusions section, all the results obtained are purely computational and, therefore, *in vitro* studies remain necessary to validate the findings.

Data availability

All the data and scripts used throughout the project can be found in the author's GitHub.

Acknowledgements

Molecular graphics and analyses performed with UCSF ChimeraX, developed by the Resource for Biocomputing, Visualization, and Informatics at the University of California, San Francisco, with support from National Institutes of Health R01-GM129325 and the Office of Cyber Infrastructure and Computational Biology, National Institute of Allergy and Infectious Diseases.

The Python and Bash scripts used throughout the project were developed with the aid of Claude Sonnet 4.6 (Anthropic). These were carefully reviewed and validated before use to ensure result accuracy. Everything in this manuscript was written by the author.

Bibliography

(1) Guan, Z.-B.; Luo, Q.; Wang, H.-R.; Chen, Y.; Liao, X.-R. Bacterial Laccases: Promising Biological Green Tools for Industrial Applications. *Cell. Mol. Life Sci. CMLS* **2018**, *75* (19), 3569–3592. https://doi.org/10.1007/s00018-018-2883-z

(2) Arregui, L.; Ayala, M.; Gómez-Gil, X.; Gutiérrez-Soto, G.; Hernández-Luna, C. E.; Herrera de los Santos, M.; Levin, L.; Rojo-Domínguez, A.; Romero-Martínez, D.; Saparrat, M. C. N.; Trujillo-Roldán, M. A.; Valdez-Cruz, N. A. Laccases: Structure, Function, and Potential Application in Water Bioremediation. *Microb. Cell Factories* **2019**, *18* (1), 200. https://doi.org/10.1186/s12934-019-1248-0

(3) Hakulinen, N.; Rouvinen, J. Three-Dimensional Structures of Laccases. *Cell. Mol. Life Sci. CMLS* **2015**, *72* (5), 857–868. https://doi.org/10.1007/s00018-014-1827-5

(4) Strong, P. J.; Claus, H. Laccase: A Review of Its Past and Its Future in Bioremediation. *Crit. Rev. Environ. Sci. Technol.* **2011**, *41* (4), 373–434. https://doi.org/10.1080/10643380902945706

(5) Mistry, J.; Chuguransky, S.; Williams, L.; Qureshi, M.; Salazar, G.; Sonnhammer, E. L. L.; Tosatto, S. C. E.; Paladin, L.; Raj, S.; Richardson, L. J.; Finn, R. D.; Bateman, A. Pfam: The Protein Families Database in 2021. *Nucleic Acids Research* **2020**, *49* (D1), D412–D419. https://doi.org/10.1093/nar/gkaa913

(6) The UniProt Consortium. UniProt: The Universal Protein Knowledgebase in 2025. *Nucleic Acids Res.* **2025**, *53* (D1), D609–D617. https://doi.org/10.1093/nar/gkae1010

(7) Jiao, M.; Chen, W.; Jiang, X.; Huang, D.; Jiang, Y.; Liu, H.; Yuan, H.; Wang, T. Recent Advances in Laccase Production: Challenges and Future Perspectives. *J. Microbiol. Biotechnol.* **2025**, *35*, e2507052. https://doi.org/10.4014/jmb.2507.07052

(8) Wu, P.; Hu, T.; Sun, L.; Fan, J. Efficient Degradation of Sulfadiazine by Photosynthetic Cyanobacteria *Synechocystis* Sp. PCC 6803 Coupling with Recombinant Laccase Strategy. *Chem. Eng. J.* **2023**, *474*, 145974. https://doi.org/10.1016/j.cej.2023.145974

(9) Xu, K.-Z.; Ma, H.; Wang, Y.-J.; Cai, Y.-J.; Liao, X.-R.; Guan, Z.-B. Extracellular Expression of Mutant CotA-Laccase SF in *Escherichia Coli* and Its Degradation of Malachite Green. *Ecotoxicol. Environ. Saf.* **2020**, *193*, 110335. https://doi.org/10.1016/j.ecoenv.2020.110335

(10) Sarker, A.; Nandi, R.; Kim, J.-E.; Islam, T. Remediation of Chemical Pesticides from Contaminated Sites through Potential Microorganisms and Their Functional Enzymes: Prospects and Challenges. *Environ. Technol. Innov.* **2021**, *23*, 101777. https://doi.org/10.1016/j.eti.2021.101777

(11) Zerva, A.; Simić, S.; Topakas, E.; Nikodinovic-Runic, J. Applications of Microbial Laccases: Patent Review of the Past Decade (2009–2019). *Catalysts* **2019**, *9* (12), 1023. https://doi.org/10.3390/catal9121023

(12) Paul, C.; Pal, N.; Maitra, M.; Das, N. Laccase-Assisted Bioremediation of Pesticides: Scope and Challenges. *Mini-Rev. Org. Chem.* **2024**, *21* (6), 633–654. https://doi.org/10.2174/1570193X20666221117161033

(13) Othman, A. M.; Flaifil, A. G. Characterization and Evaluation of the Immobilized Laccase Enzyme Potential in Dye Degradation via One Factor and Response Surface Methodology Approaches. *Sci. Rep.* **2025**, *15* (1), 735. https://doi.org/10.1038/s41598-024-82310-0

(14) Kyomuhimbo, H. D.; Brink, H. G. Applications and Immobilization Strategies of the Copper-Centred Laccase Enzyme; a Review. *Heliyon* **2023**, *9* (2), e13156. https://doi.org/10.1016/j.heliyon.2023.e13156

(15) Yamaguchi, H.; Miyazaki, M. Bioremediation of Hazardous Pollutants Using Enzyme-Immobilized Reactors. *Molecules* **2024**, *29* (9), 2021. https://doi.org/10.3390/molecules29092021
(16) Tiwari, A.; Krisnawati, D. I.; Widodo; Cheng, T.-M.; Kuo, T.-R. Precision Thermostability Predictions: Leveraging Machine Learning for Examining Laccases and Their Associated Genes. *Int. J. Mol. Sci.* **2024**, *25* (23), 13035. https://doi.org/10.3390/ijms252313035
(17) Tien, M. Z.; Meyer, A. G.; Sydykova, D. K.; Spielman, S. J.; Wilke, C. O. Maximum Allowed Solvent Accessibilites of Residues in Proteins. *PLoS ONE* **2013**, *8* (11), e80635. https://doi.org/10.1371/journal.pone.0080635
(18) Chen, H.; Zhou, H.-X. Prediction of Solvent Accessibility and Sites of Deleterious Mutations from Protein Sequence. *Nucleic Acids Res.* **2005**, *33* (10), 3193–3199. https://doi.org/10.1093/nar/gki633
(19) Pezeshgi Modarres, H.; R. Mofrad, M.; Sanati-Nezhad, A. Protein Thermostability Engineering. *RSC Adv.* **2016**, *6* (116), 115252–115270. https://doi.org/10.1039/C6RA16992A
(20) Baldwin, R. L.; Rose, G. D. How the Hydrophobic Factor Drives Protein Folding. *Proc. Natl. Acad. Sci.* **2016**, *113* (44), 12462–12466. https://doi.org/10.1073/pnas.1610541113
(21) Zhu, W.; Liu, Y.; Cao, H.; Liu, L.; Tan, T. Short-Loop Engineering Strategy for Enhancing Enzyme Thermal Stability. *iScience* **2025**, *28* (4), 112202. https://doi.org/10.1016/j.isci.2025.112202
(22) Peccati, F.; Segovia, C. M.; Núñez-Franco, R.; Jiménez-Osés, G. Computation of Protein Thermostability and Epistasis. *WIREs Comput. Mol. Sci.* **2025**, *15* (5), e70045. https://doi.org/10.1002/wcms.70045
(23) Miotto, M.; Olimpieri, P. P.; Di Rienzo, L.; Ambrosetti, F.; Corsi, P.; Lepore, R.; Tartaglia, G. G.; Milanetti, E. Insights on Protein Thermal Stability: A Graph Representation of Molecular Interactions. *Bioinformatics* **2018**, *35* (15), 2569–2577. https://doi.org/10.1093/bioinformatics/bty1011
(24) Sayers, E. W.; Beck, J.; Bolton, E. E.; Brister, J. R.; Chan, J.; Connor, R.; Feldgarden, M.; Fine, A. M.; Funk, K.; Hoffman, J.; Kannan, S.; Kelly, C.; Klimke, W.; Kim, S.; Lathrop, S.; Marchler-Bauer, A.; Murphy, T. D.; O'Sullivan, C.; Schmieder, E.; Skripchenko, Y.; Stine, A.; Thibaud-Nissen, F.; Wang, J.; Ye, J.; Zellers, E.; Schneider, V. A.; Pruitt, K. D. Database Resources of the National Center for Biotechnology Information in 2025. *Nucleic Acids Res.* **2024**, *53* (D1), D20–D29. https://doi.org/10.1093/nar/gkae979
(25) Hauenstein, J.; Jeske, L.; Jäde, A.; Krull, M.; Dümmer, K.; Koblitz, J.; Tietz, A.; Jahn, D.; Reimer, L. C.; Bunk, B. BRENDA in 2026: A Global Core Biodata Resource for Functional Enzyme and Metabolic Data within the DSMZ Digital Diversity. *Nucleic Acids Res.* **2025**, *54* (D1), D527–D534. https://doi.org/10.1093/nar/gkaf1113
(26) Rahban, M.; Zolghadri, S.; Salehi, N.; Ahmad, F.; Haertlé, T.; Rezaei-Ghaleh, N.; Sawyer, L.; Saboury, A. A. Thermal Stability Enhancement: Fundamental Concepts of Protein Engineering Strategies to Manipulate the Flexible Structure. *Int. J. Biol. Macromol.* **2022**, *214*, 642–654. https://doi.org/10.1016/j.ijbiomac.2022.06.154
(27) Fu, L.; Niu, B.; Zhu, Z.; Wu, S.; Li, W. CD-HIT: Accelerated for Clustering the next-Generation Sequencing Data. *Bioinformatics* **2012**, *28* (23), 3150–3152. https://doi.org/10.1093/bioinformatics/bts565
(28) Katoh, K.; Rozewicki, J.; Yamada, K. D. MAFFT Online Service: Multiple Sequence Alignment, Interactive Sequence Choice and Visualization. *Brief. Bioinform.* **2017**, *20* (4), 1160–1166. https://doi.org/10.1093/bib/bbx108

(29) Capella-Gutierrez, S.; Silla-Martinez, J. M.; Gabaldon, T. TrimAl: A Tool for Automated Alignment Trimming in Large-Scale Phylogenetic Analyses. *Bioinformatics* **2009**, *25* (15), 1972–1973. https://doi.org/10.1093/bioinformatics/btp348

(30) Enguita, F. J.; Martins, L. O.; Henriques, A. O.; Carrondo, M. A. Crystal Structure of a Bacterial Endospore Coat Component: A LACCASE WITH ENHANCED THERMOSTABILITY PROPERTIES*. *J. Biol. Chem.* **2003**, *278* (21), 19416–19425. https://doi.org/10.1074/jbc.M301251200

(31) Berman, H.; Henrick, K.; Nakamura, H. Announcing the Worldwide Protein Data Bank. *Nat. Struct. Mol. Biol.* **2003**, *10* (12), 980–980. https://doi.org/10.1038/nsb1203-980

(32) Mitternacht, S. FreeSASA: An Open Source C Library for Solvent Accessible Surface Area Calculations. F1000Research February 18, 2016. https://doi.org/10.12688/f1000research.7931.1

(33) Pettersen, E. F.; Goddard, T. D.; Huang, C. C.; Meng, E. C.; Couch, G. S.; Croll, T. I.; Morris, J. H.; Ferrin, T. E. UCSF ChimeraX: Structure Visualization for Researchers, Educators, and Developers. *Protein Sci. Publ. Protein Soc.* **2021**, *30* (1), 70–82. https://doi.org/10.1002/pro.3943

(34) Lemoine, F.; Gascuel, O. Gotree/Goalign: Toolkit and Go API to Facilitate the Development of Phylogenetic Workflows. *NAR Genomics Bioinforma.* **2021**, *3* (3), lqab075. https://doi.org/10.1093/nargab/lqab075

(35) Waterhouse, A. M.; Procter, J. B.; Martin, D. M. A.; Clamp, M.; Barton, G. J. Jalview Version 2—a Multiple Sequence Alignment Editor and Analysis Workbench. *Bioinformatics* **2009**, *25* (9), 1189–1191. https://doi.org/10.1093/bioinformatics/btp033

(36) *Anthropic*. Claude Sonnet 4.6. https://claude.com/product/overview (accessed 2026-05-12)

(37) Virtanen, P.; Gommers, R.; Oliphant, T. E.; Haberland, M.; Reddy, T.; Cournapeau, D.; Burovski, E.; Peterson, P.; Weckesser, W.; Bright, J.; van der Walt, S. J.; Brett, M.; Wilson, J.; Millman, K. J.; Mayorov, N.; Nelson, A. R. J.; Jones, E.; Kern, R.; Larson, E.; Carey, C. J.; Polat, İ.; Feng, Y.; Moore, E. W.; VanderPlas, J.; Laxalde, D.; Perktold, J.; Cimrman, R.; Henriksen, I.; Quintero, E. A.; Harris, C. R.; Archibald, A. M.; Ribeiro, A. H.; Pedregosa, F.; van Mulbregt, P. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nat. Methods* **2020**, *17* (3), 261–272. https://doi.org/10.1038/s41592-019-0686-2

(38) Harris, C. R.; Millman, K. J.; van der Walt, S. J.; Gommers, R.; Virtanen, P.; Cournapeau, D.; Wieser, E.; Taylor, J.; Berg, S.; Smith, N. J.; Kern, R.; Picus, M.; Hoyer, S.; van Kerkwijk, M. H.; Brett, M.; Haldane, A.; del Río, J. F.; Wiebe, M.; Peterson, P.; Gérard-Marchant, P.; Sheppard, K.; Reddy, T.; Weckesser, W.; Abbasi, H.; Gohlke, C.; Oliphant, T. E. Array Programming with NumPy. *Nature* **2020**, *585* (7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

(39) McKinney, W. Data Structures for Statistical Computing in Python; Austin, Texas, 2010; pp 56–61. https://doi.org/10.25080/Majora-92bf1922-00a

(40) Hunter, J. D. Matplotlib: A 2D Graphics Environment. *Comput. Sci. Eng.* **2007**, *9* (3), 90–95. https://doi.org/10.1109/MCSE.2007.55

(41) Pommié, C.; Levadoux, S.; Sabatier, R.; Lefranc, G.; Lefranc, M. IMGT Standardized Criteria for Statistical Analysis of Immunoglobulin V-REGION Amino Acid Properties. *J. Mol. Recognit.* **2004**, *17* (1), 17–32. https://doi.org/10.1002/jmr.647

(42) Kyte, J.; Doolittle, R. F. A Simple Method for Displaying the Hydropathic Character of a Protein. *J. Mol. Biol.* **1982**, *157* (1), 105–132. https://doi.org/10.1016/0022-2836(82)90515-0

(43) Zamyatnin, A. A. Protein Volume in Solution. *Prog. Biophys. Mol. Biol.* **1972**, *24*, 107–123. https://doi.org/10.1016/0079-6107(72)90005-3

(44) Zimmerman, J. M.; Eliezer, N.; Simha, R. The Characterization of Amino Acid Sequences in Proteins by Statistical Methods. *J. Theor. Biol.* **1968**, *21* (2), 170–201. https://doi.org/10.1016/0022-5193(68)90069-6
(45) Bhaskaran, R.; Ponnuswamy, P. K. Positional Flexibilities of Amino Acid Residues in Globular Proteins. *Int. J. Pept. Protein Res.* **1988**, *32* (4), 241–255. https://doi.org/10.1111/j.1399-3011.1988.tb01258.x
(46) Gasteiger, E.; Hoogland, C.; Gattiker, A.; Duvaud, S.; Wilkins, M. R.; Appel, R. D.; Bairoch, A. Protein Identification and Analysis Tools on the ExPASy Server. In *The Proteomics Protocols Handbook*; Walker, J. M., Ed.; Humana Press: Totowa, NJ, 2005; pp 571–607. https://doi.org/10.1385/1-59259-890-0:571
(47) Vicente, A. I.; Viña-Gonzalez, J.; Mateljak, I.; Monza, E.; Lucas, F.; Guallar, V.; Alcalde, M. Enhancing Thermostability by Modifying Flexible Surface Loops in an Evolved High-Redox Potential Laccase. *AIChE J.* **2020**, *66* (3), e16747. https://doi.org/10.1002/aic.16747
(48) Richardson, J. S.; Richardson, D. C. Natural β-Sheet Proteins Use Negative Design to Avoid Edge-to-Edge Aggregation. *Proc. Natl. Acad. Sci.* **2002**, *99* (5), 2754–2759. https://doi.org/10.1073/pnas.052706099
(49) Shamsir, M. S.; Dalby, A. R. β-Sheet Containment by Flanking Prolines: Molecular Dynamic Simulations of the Inhibition of β-Sheet Elongation by Proline Residues in Human Prion Protein. *Biophys. J.* **2007**, *92* (6), 2080–2089. https://doi.org/10.1529/biophysj.106.092320
(50) Rigo, A.; Corazza, A.; Luisa di Paolo, M.; Rossetto, M.; Ugolini, R.; Scarpa, M. Interaction of Copper with Cysteine: Stability of Cuprous Complexes and Catalytic Role of Cupric Ions in Anaerobic Thiol Oxidation. *J. Inorg. Biochem.* **2004**, *98* (9), 1495–1501. https://doi.org/10.1016/j.jinorgbio.2004.06.008
(51) Kim, T.; Joo, J. C.; Yoo, Y. J. Hydrophobic Interaction Network Analysis for Thermostabilization of a Mesophilic Xylanase. *J. Biotechnol.* **2012**, *161* (1), 49–59. https://doi.org/10.1016/j.jbiotec.2012.04.015
(52) Rhys, N. H.; Soper, A. K.; Dougan, L. The Hydrogen-Bonding Ability of the Amino Acid Glutamine Revealed by Neutron Diffraction Experiments. *J. Phys. Chem. B* **2012**, *116* (45), 13308–13319. https://doi.org/10.1021/jp307442f